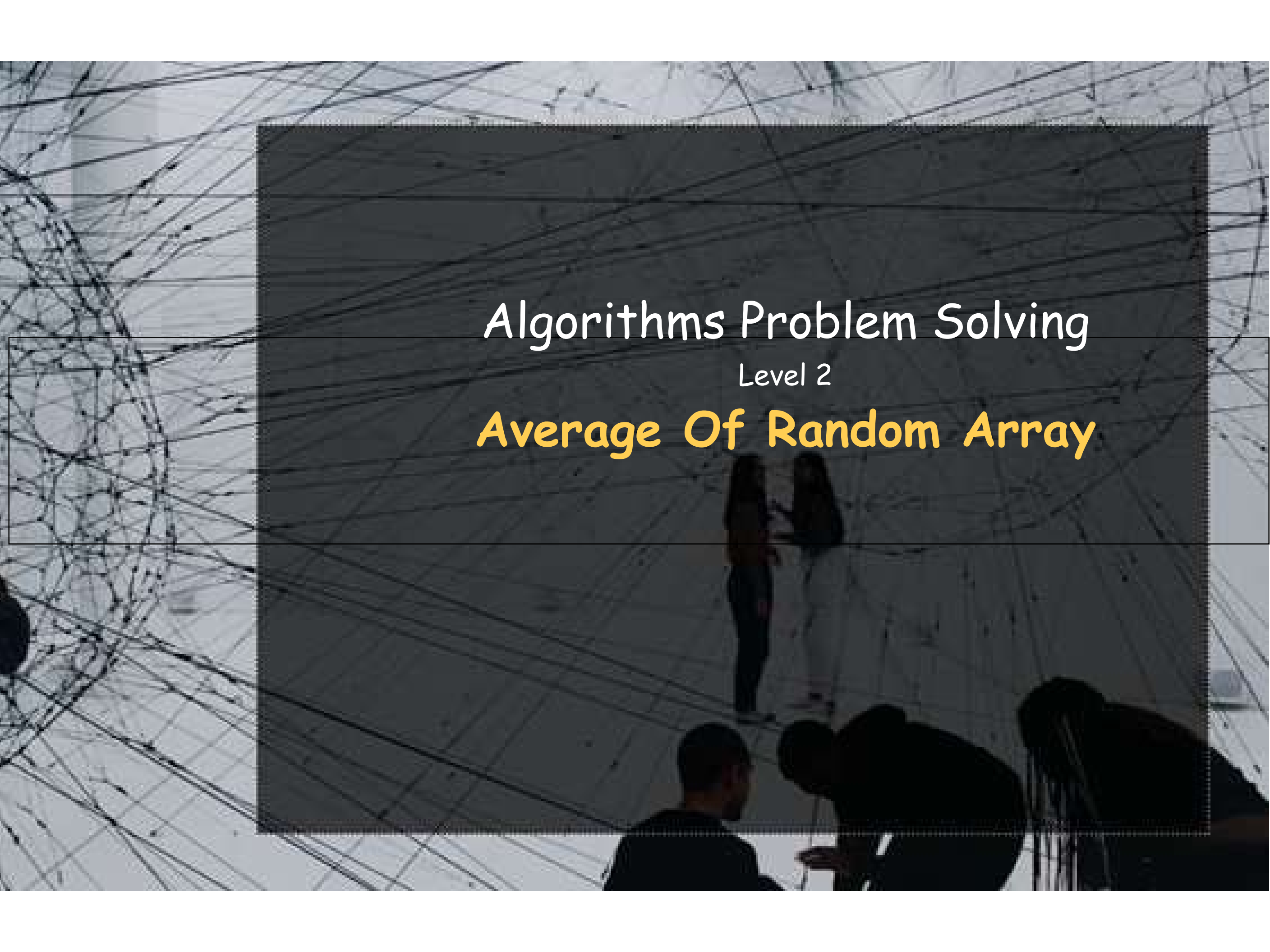


Algorithms Problem Solving

Level 2

Average Of Random Array



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Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It then calculates the average of all array elements.
- ◆ The average is calculated by dividing the sum of all elements by the number of elements.
- ◆ The formula is: $\text{Average} = \text{Sum} / \text{Number of Elements}$.

Explanation

- ◆ ``FillArrayWithRandomNumber()`` fills the array with random numbers from ``1`` to ``100``.
- ◆ ``PrintArrayElements()`` prints all elements.
- ◆ ``SumAllNumbInArray()`` is intended to calculate the total sum.
- ◆ ``AverageOfArray()`` calls ``SumAllNumbInArray()`` and divides the result by ``length``.
- ◆ However, the ``SumAllNumbInArray()`` function in the given code is incorrect, so the average will also be incorrect.
- ◆ The sum function should add every element, not only elements smaller than the current sum.

Steps in main()

1. Ask the user for the number of elements.
2. Fill the array with random numbers from `1` to `100`.
3. Print all array elements.
4. Calculate the sum of all elements.
5. Divide the sum by the number of elements.
6. Return the average as a `double`.
7. Print the average.

💡 Example

◆ Number of elements: `10`

◆ Array:

42 68 35 1 70 25 79 59 63 65

◆ Sum:

$42 + 68 + 35 + 1 + 70 + 25 + 79 + 59 + 63 + 65 = 507$

◆ Average:

$507 / 10 = 50.7$

➡ Result

Array Elements: 42 68 35 1 70 25 79 59 63 65

Average of all Numbers is: 50.7

◆ The example output in the comment says `43`, but for the listed array the correct average is `50.7`.

✚ Note

- ◆ The `SumAllNumbInArray()` function should be corrected to:

```
int SumAllNumbInArray(int arr[], int length)
{
    int sum = 0;
    for (int i = 0; i < length; i++)
    {
        sum += arr[i];
    }
    return sum;
}
```

- ◆ Then `AverageOfArray()` can calculate:

```
double AverageOfArray(int arr[], int length)
{
    return (double)SumAllNumbInArray(arr, length) / length;
}
```

- ◆ The `(double)` ensures floating-point division.

- ◆ Without it, if both `sum` and `length` are integers, C++ performs integer division before returning the result as `double`.

⊘ Warning

- ◆ The current `SumAllNumbInArray()` is incorrect:

```
int sum = arr[0];
for (int i = 0; i < length; i++)
{
    if (arr[i] < sum)
        sum += arr[i];
}
```

- ◆ It does not calculate the total sum.
- ◆ Because the sum is wrong, the average is also wrong.
- ◆ `ReadArrayElements()` is defined but never used.
- ◆ `cmath` and `corecrt\_math\_defines.h` are unnecessary here.
- ◆ `arrLength` should ideally be validated so it does not exceed `100`.
- ◆ `rand()` is not seeded with `srand()`, so the same pseudo-random sequence may be produced between runs.



Comparison

▨ `SumAllNumbInArray()`

- ✓ Calculates the total of all array elements.
- ✓ Adds every element using `sum += arr[i]`.
- ✓ Example: `10 20 30` → `60`.

▤ `AverageOfArray()`

- ✓ Calculates the average of all array elements.
- ✓ Uses the sum and divides it by `length`.
- ✓ Example: `10 20 30` → `20`.
- ✓ Returns a `double` to support decimal results.

✦ Summary

- ✓ Fill the array with random numbers.
- ✓ Calculate the total sum of the elements.
- ✓ Divide the sum by the number of elements.
- ✓ Use `double` for the average when decimal values are possible.
- ✓ Cast the sum to `double` before division to avoid integer division.

Thank you 😊